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Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

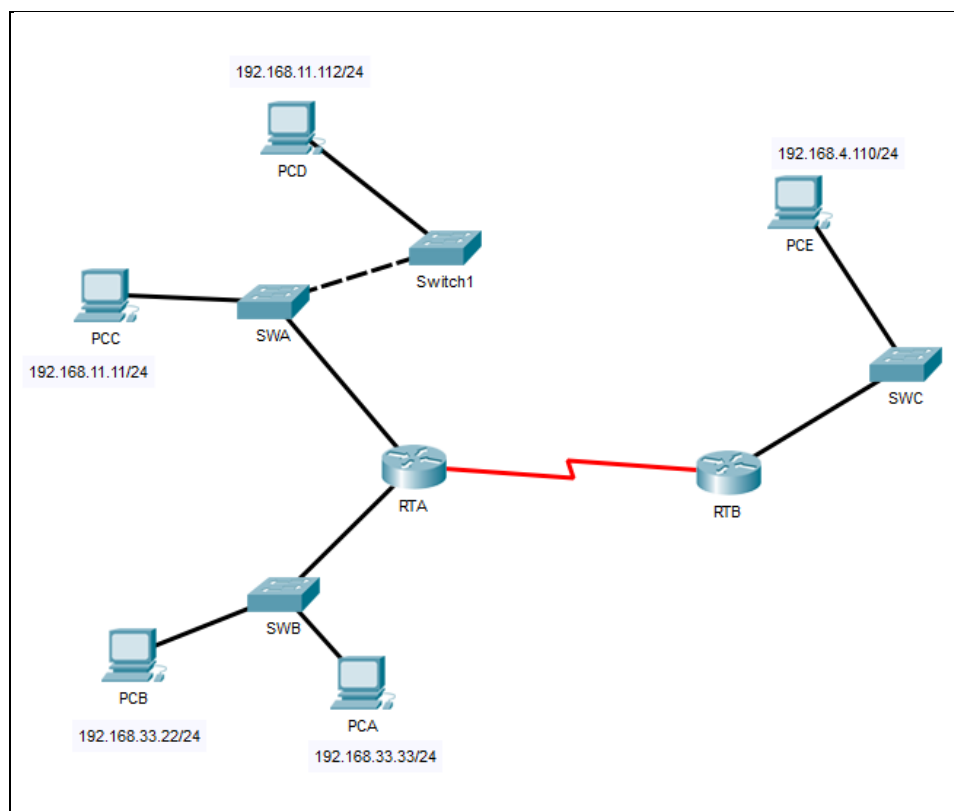


Figure 1

Part 1: Review the topology

Step 1: Open the *Lab_4_ARP.pkt*, and Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.4.1       -          0001.977A.B614  ARPA   FastEthernet0/0
---
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

SWA

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0002.4a00.0e91    DYNAMIC     Fa0/1
1       000c.8546.7d85    DYNAMIC     Fa1/1
```

SWB

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       000c.cf0c.593a    DYNAMIC     Fa0/1
```

SWC

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0001.977a.b614    DYNAMIC     Fa0/1
```

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp -a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

PCA

```
C:\>arp -a
No ARP Entries Found
C:\>|
```

PCB

```
C:\>arp -a
No ARP Entries Found
C:\>|
```

PCC

```
C:\>arp -a
No ARP Entries Found
C:\>|
```

PCD

```
C:\>arp -a
No ARP Entries Found
C:\>|
```

PCE

```
C:\>arp -a
No ARP Entries Found
C:\>|
```

e. What are your thoughts on the results?

No ARP Entries Found because MAC addresses aren't assigned to each PC.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time=2ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time=13ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

- In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746        dynamic
```

- In the Command prompt of PCB, type **arp -a**. Paste the result of this command here

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.33.33         0002.1755.9a06        dynamic
```

- In the Command prompt of PCC, PCD abd PCE, type **arp -a**. Paste the result of this command here.

PCC

```
C:\>arp -a
No ARP Entries Found
C:\>|
```

PCD

```
C:\>arp -a
No ARP Entries Found
C:\>|
```

PCE

```
C:\>arp -a
No ARP Entries Found
C:\>
```

Step 2: Generate traffic between PCC to all other PC except PCA.

- a. Click **PCC** and click the Desktop tab > Command Prompt.
- b. Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```
Pinging 192.168.33.22 with 32 bytes of data:

Request timed out.
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
```

- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time=1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=7ms TTL=126
Reply from 192.168.4.110: bytes=32 time=9ms TTL=126
Reply from 192.168.4.110: bytes=32 time=8ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 9ms, Average = 8ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
192.168.11.112        0001.6462.0278        dynamic
```

- e. Discuss the results you got from all the commands on PCC.
- When PCC pings PCD and PCN, the PC now has MAC addresses but when PCC tried to ping PCE, which from a different router and subnet it fails to resolve MAC address because PCE has a different subnet.
- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address      Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1  -         0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.11.11 3         00D0.D39A.C0D9  ARPA   FastEthernet1/0
Internet  192.168.33.1  -         000C.CF0C.593A  ARPA   FastEthernet0/0
Internet  192.168.33.22 3         0060.47EA.A746  ARPA   FastEthernet0/0
---
```

- g. At Router RTA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.4.1      -         0001.977A.B614  ARPA   FastEthernet0/0
Internet  192.168.4.110    2         0060.702D.7C08  ARPA   FastEthernet0/0
```

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1     0001.6462.0278   DYNAMIC   Fa1/1
1     0002.4a00.0e91   DYNAMIC   Fa0/1
1     000c.8546.7d85   DYNAMIC   Fa1/1
1     00d0.d39a.c0d9   DYNAMIC   Fa2/1
```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    000c.cf0c.593a    DYNAMIC Fa0/1
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0001.977a.b614    DYNAMIC Fa0/1
-----"
```

d. Do switches use arp table? (Y/N)

No

e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

Switches does not use arp table because ARP operates at layer 3 but switches operate at level 2.

f. What information does the command `show mac-address-table` gives?

MAC addresses table, the information given is MAC addresses, VLAN, port, Type

Part 3: Attach wireless lab results.

In this part, you will use **Lab_4_Wireless.pka** file.

Step1: Change the filename of the pka file.

- Change the Lab 4 filename to include your name. **Example: Lab4AliAhmad.pkt*
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot displays the Packet Tracer interface. On the left, a network topology is shown with a switch (S1) connected to a router (R1) and a wireless router (WRS2). PC1 is connected to S1, PC2 to WRS2, and PC3 to R1. IP addresses and VLANs are assigned to each device. On the right, the 'Packet Tracer - Configuring Wireless LAN Access' activity wizard is open. It shows an 'Addressing Table' with the following data:

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

At the bottom of the wizard, the 'Completion' status is shown as 100/100, indicating that the configuration is complete.

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

Cisco Packet Tracer - C:/Users/dania/Downloads/Lab_4_Wireless.pka - SyarifahDania - 2025-01-30 22:20:51

File Edit Options View Tools Extensions Window Help

Logical Physical x 357, y 14

Subinterfaces

- G0/0/10 172.17.10.1/24
- G0/0/20 172.17.20.1/24
- G0/0/88 172.17.88.1/24

PC1 172.17.10.21/24 VLAN 10

PC2 172.17.20.22/24 VLAN 20

WRS2 172.17.88.25/24 VLAN 88 172.17.40.100/24

PT Activity: 00:08:45

Part 1: Configure a Wireless Router

Step 1: Connect the Internet interface of WRS2 to S1.
Connect the WRS2 Internet interface to the S1 F0/7 interface.

Step 2: Configure the Internet connection type.

- Click WRS2 > GUI tab.
- Set the Internet Connection type to Static IP.
- Configure the IP addressing according to the Addressing Table.

Step 3: Configure the network setup.

- Scroll down to **Network Setup**. For the Router IP option, set the IP address to 172.17.40.1 and the subnet mask to 255.255.255.0.
- Enable the DHCP server.
- Scroll to the bottom of the page and click **Save Settings**.

Step 4: Configure wireless access and security.

- At the top of the window, click **Wireless**. Set the **Network Mode** to **Wireless-N Only** and change the SSID to **WRS_LAN**.
- Disable **SSID Broadcast** and click **Save Settings**.
- Click the **Wireless Security** option.

Time Elapsed: 00:08:45 Completion: 100/100

☐ Top ☐ Dock 1/1

PC3

Link Information

Infrastructure Mode 2.4GHz

You have successfully connected to the access point

Signal Strength Link Quality Adaptor is Active

Wireless-N Notebook Adapter Wireless Network Monitor v1.0 Model No. WPC300N

Color	Time(sec)	Periodic	Num	Edit	Delete
	0.000	N	0	(edit)	
	0.000	N	1	(edit)	

Cooper Straight-Through